MSC 201 **26077**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For comparison, sample SD and confidence are held fixed while n increases. What happens to a t mean interval's margin?	A. It stays unchanged. B. It increases because there are more observations. C. It decreases as SE and the critical t value decrease. D. It becomes the population SD. 10 pts
2. Which change best addresses a poll that includes only frequent library users?	A. Broaden selection to include nonusers. B. Shuffle the existing responses. C. Collect ten times as many frequent-user responses. D. Report more decimal places. 10 pts
3. A service meets its deadline with probability 0.80. What is the probability of missing it?	A. 0.80. B. 0.20. C. 0.40. D. 1.80. 10 pts
4. Using the reported quartiles, group A has Q1=3, Q3=9; group B has Q1=10, Q3=14. Which has greater IQR?	A. The larger median must determine the IQR. B. They are equal. C. Group B. D. Group A. 10 pts
5. A symmetric confidence interval is 8 to 12. What point estimate is at its center?	A. 10. B. 8. C. 4. D. 12. 10 pts
6. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 99.7%. B. 3%. C. 0.15%. D. 0.3%. 10 pts

Test continues on the other side of the sheet.

MSC 201 **26077**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

7. The sample is an unordered set of cables. Why count combinations rather than different orders of the same set?	A. Reordering the same selected cables does not create a new sample set. B. Selection order determines whether a cable is faulty. C. Every sample set has only one cable. D. Combinations assume replacement. 10 pts
8. A game pays a prize of 8 and costs 3 to play. What is the net gain on that outcome?	A. 5. B. 3. C. 8. D. 11. 10 pts
9. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The observations must include negative waits. B. The true sample mean can actually be below every observation. C. It is approximation error outside the true support. D. The normal model is exact at every tail. 10 pts
10. An exponential model has mean wait 4 minutes. What is its SD?	A. 4 minutes. B. 16 minutes. C. 0 minutes. D. 2 minutes. 10 pts

26077

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5. A.

10 pts

6.

D.

10 pts

TEST PAGE 2

7. A.

10 pts

8. A.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **21345**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A late-job rate is 10% overall but 30% during storms. What does storm information do?	A. It changes the estimated chance of lateness. B. It establishes that every storm job is late. C. It proves the events are independent. D. It makes lateness impossible outside storms. 10 pts
2. The mean waiting count is 4 trials. Must the first success occur by trial 4?	A. Yes; the mean is a deadline. B. No; it must occur exactly on trial 5. C. No; longer waits remain possible. D. Yes; 4 is the maximum. 10 pts
3. 400 people are sampled without replacement from 20,000. Does this meet the at-most-10% sample-fraction guideline?	A. No; n must be below 30. B. Yes; any random sample meets it. C. Yes; the fraction is 2%. D. No; the fraction is 20%. 10 pts
4. A frequency table lists 1 page for 8 jobs and 5 pages for 2 jobs. Why is (1+5)/2 inappropriate for mean pages per job?	A. It gives extra weight to 1-page jobs. B. It ignores how often each value occurs. C. It counts too many individual jobs. D. It is always the median. 10 pts
5. The course's binomial normal approximation check requires np>5 and n(1-p)>5. For n=100, p=0.02, which judgment fits?	A. It fails because expected failures are 98. B. It fails because expected successes are only 2. C. It passes because p is positive. D. It passes because n exceeds 30. 10 pts
6. Two datasets have identical five-number summaries. What follows?	A. They can still have different internal distributions. B. Their means must be identical. C. Their sample sizes must be identical. D. They must contain identical observations. 10 pts
7. A value equals its distribution's mean. What is its z-score?	A. 0. B. The mean. C. The SD. D. 1. 10 pts

Test continues on the other side of the sheet.

MSC 201 **21345**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Mean is 100, SD is 10, and the supplied 90th-percentile z is 1.28. What is the corresponding value?	A. 87.2. B. 12.8. C. 112.8. D. 101.28. 10 pts
9. A continuous uniform model runs from 4 to 12. What is its mean?	A. 4. B. 16. C. 8. D. 12. 10 pts
10. A random sample of 80 enrolled students reports 30% support for later hours. Which conclusion is justified?	A. 30% describes the sample and estimates population support. B. The population percentage must exceed 30%. C. Exactly 24 students in the entire college support later hours. D. Exactly 30% of all students support later hours. 10 pts

21345

TEST PAGE 1

1. A.

10 pts

2. C.

10 pts

3. C.

10 pts

4. B.

10 pts

5. B.

10 pts

6. A.

10 pts

7. A.

10 pts

TEST PAGE 2

8. C.

10 pts

9. C.

10 pts

10. A.

10 pts

MSC 201 **26290**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Two normal curves share SD but have different means. What is equal?	A. Their 90th-percentile values. B. Their centers. C. Their widths and shapes. D. Their probabilities below every fixed cutoff. 10 pts
2. How does no arrivals in the next 5 minutes relate to waiting more than 5 minutes for the next arrival?	A. The first requires exactly one arrival. B. They are complements. C. They are independent unrelated events. D. They describe the same event, ignoring a zero-probability boundary. 10 pts
3. Why does a t interval use a larger critical multiplier than z at small degrees of freedom?	A. The confidence level is automatically 100%. B. The observations must all be larger. C. Estimating SD adds uncertainty reflected in heavier tails. D. The population mean is random in the formula. 10 pts
4. A bar graph of counts 98 and 100 starts its vertical axis at 97. What impression might it exaggerate?	A. The number of categories. B. The total number of observations. C. The fact that 98 is below 100. D. The size of the difference between the counts. 10 pts
5. Which quantity describes how sample means vary from sample to sample?	A. The sample size. B. The population mean. C. The range of all individuals alone. D. Standard error. 10 pts
6. In 2, 3, 3, 4, 18, the 18 is replaced by 180. What happens?	A. The median increases more than the mean. B. The mean decreases. C. Both stay unchanged. D. The mean increases; the median stays 3. 10 pts
7. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It decreases. B. It stays unchanged. C. It increases. D. It becomes zero. 10 pts

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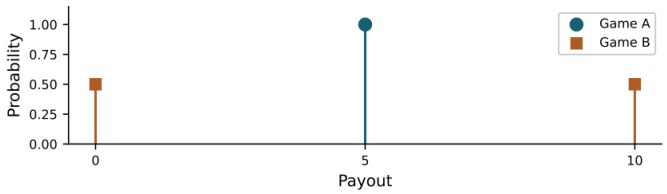
MSC 201 **26290**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A spinner has three colors occupying half, one quarter, and one quarter of its area. Why is assigning each color probability 1/3 inappropriate?	A. Every color must have probability 1/2. B. Probabilities must be whole numbers. C. A spinner cannot have three outcomes. D. The color areas are unequal. 10 pts
9. Game A pays 5 with certainty. Game B pays 0 or 10 with equal chance. What is true? 	A. Both have zero spread. B. A has greater spread. C. B has a larger expected payout. D. They have equal expected payout; B has greater spread. 10 pts
10. There are 20 independent trials with p=0.3. What is the expected success count?	A. 14. B. 6. C. 0.3. D. 20. 10 pts

26290

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

D.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

D.

10 pts

10.

B.

10 pts

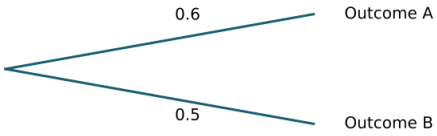
MSC 201 **25641**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Ten cables are sampled without replacement from 100,000. Why might binomial be a useful approximation?	A. The small sample barely changes the population composition. B. The faulty count is continuous. C. No-replacement draws become exactly independent. D. All samples have the same faulty count. 10 pts
2. A population has mean 50. Where are unbiased random-sample means centered?	A. At zero. B. At 50. C. At the population SD. D. At 50 divided by sample size. 10 pts
3. Two reasonable quartile methods give upper outlier fences of 18 and 20 for the same dataset. One observation is 35. Which conclusion agrees with both methods?	A. The observation is not flagged under either method. B. Only the method with fence 18 flags the observation. C. The observation must be a recording error. D. The observation is flagged as unusually high under both methods. 10 pts
4. At a tree node, two exhaustive outgoing branch probabilities are 0.6 and 0.5. What is wrong? 	A. They total more than 1. B. All tree branches must be 0.5. C. They should both be zero. D. Branches cannot represent conditional probabilities. 10 pts
5. Both endpoints of a uniform model increase by 4. What happens to its mean and SD?	A. Both stay unchanged. B. Both rise by 4. C. SD rises by 4; mean stays unchanged. D. Mean rises by 4; SD stays unchanged. 10 pts
6. A model predicts 5% of items fall outside specifications. Among 200 items, how many are expected outside?	A. 5. B. 10. C. 40. D. Exactly 10 must occur. 10 pts

Test continues on the other side of the sheet.

MSC 201 **25641**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A count variable has expected value 0.4. What does that mean?	A. A single session must contain 0.4 events. B. Exactly 40% of sessions must contain one event. C. The model is invalid. D. Its long-run average can be 0.4 despite integer observations. 10 pts
8. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A proportion interval regardless of variable type. B. A Poisson count probability. C. A known-SD z interval treating s as known population SD. D. A one-sample t interval. 10 pts
9. A printer study uses only one printer per paper type. Why is that weak evidence?	A. Two observations guarantee a representative result. B. Replication would eliminate all possible errors. C. Paper type cannot affect printer performance. D. Printer-specific differences may dominate the comparison. 10 pts
10. One hundred valid 95% intervals are constructed in repeated sampling. Must exactly 95 cover the mean?	A. Yes; the confidence level fixes that count. B. Yes; any other count proves bias. C. No; the coverage count itself can vary. D. No; all 100 must cover it. 10 pts

25641

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. D.

10 pts

4. A.

10 pts

5. D.

10 pts

6. B.

10 pts

TEST PAGE 2

7. D.

10 pts

8. D.

10 pts

9. D.

10 pts

10. C.

10 pts

MSC 201 **90916**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A researcher takes 1,000 samples of 25 packages and plots each sample mean. What does one plotted value represent?	A. The mass of one package. B. A population parameter that changes on each sample. C. The mean of one 25-package sample. D. The mean of all 25,000 packages combined. 10 pts
2. A spinner has three colors occupying half, one quarter, and one quarter of its area. Why is assigning each color probability 1/3 inappropriate?	A. A spinner cannot have three outcomes. B. Every color must have probability 1/2. C. The color areas are unequal. D. Probabilities must be whole numbers. 10 pts
3. A normal model has mean 20. What is P(X<20)?	A. 0. B. The SD. C. 0.5. D. 1. 10 pts
4. The sample is an unordered set of cables. Why count combinations rather than different orders of the same set?	A. Every sample set has only one cable. B. Selection order determines whether a cable is faulty. C. Combinations assume replacement. D. Reordering the same selected cables does not create a new sample set. 10 pts
5. For a continuous duration, how do P(X<6) and P(X<=6) compare?	A. They are equal. B. They differ by the density height at 6. C. The second is always twice the first. D. The first is always zero. 10 pts
6. A rate estimate rises from 40% to 45%. What is its increase in percentage points?	A. 0.05 percentage points. B. 5 percentage points. C. 12.5 percentage points. D. 45 percentage points. 10 pts

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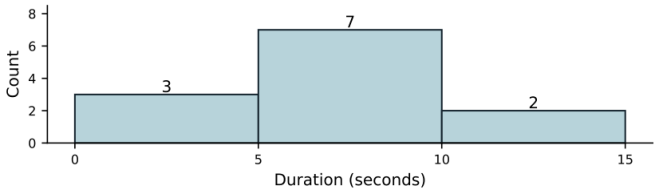
MSC 201 **90916**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Remove 100 from 1, 2, 3, 4, 100. What happens to the mean and median?	A. The median increases and mean stays unchanged. B. Both decrease, with a much larger decrease in the mean. C. The mean increases and median decreases. D. Both stay unchanged. 10 pts
8. A symmetric confidence interval is 8 to 12. What point estimate is at its center?	A. 10. B. 8. C. 12. D. 4. 10 pts
9. Histogram counts are 3 in [0,5), 7 in [5,10), and 2 in [10,15). How many values are below 10? 	A. 7. B. 2. C. 12. D. 10. 10 pts
10. A rare-event binomial count has n=1,000 and p=0.002. What mean would its Poisson approximation use?	A. 2. B. 500. C. 0.002. D. 1,000. 10 pts

90916

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5. A.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

B.

10 pts

8. A.

10 pts

9.

D.

10 pts

10. A.

10 pts

MSC 201 **25376**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A is 6 above its mean with SD 3; B is 8 above its mean with SD 8. Which is more extreme above its group center?	A. The larger raw difference always determines extremeness. B. A. C. They are equally extreme. D. B. 10 pts
2. Why does a t interval use a larger critical multiplier than z at small degrees of freedom?	A. The observations must all be larger. B. The confidence level is automatically 100%. C. Estimating SD adds uncertainty reflected in heavier tails. D. The population mean is random in the formula. 10 pts
3. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. Success probability exceeds 0.5. B. Every trial must succeed. C. The expected count is an integer. D. Fixed trial count, two outcomes, independence, and constant success chance. 10 pts
4. A proposed event probability is 1.15. What is wrong?	A. Probabilities must always equal 0.5. B. All probabilities must be integers. C. A probability cannot exceed 1. D. Only complements may exceed 1. 10 pts
5. A count variable has expected value 0.4. What does that mean?	A. Its long-run average can be 0.4 despite integer observations. B. The model is invalid. C. A single session must contain 0.4 events. D. Exactly 40% of sessions must contain one event. 10 pts
6. Students are randomly invited, but only 15% respond. Which concern remains?	A. Random invitation guarantees representative responses. B. The invitations were necessarily nonrandom. C. Low response proves every answer false. D. Respondents may differ from nonrespondents. 10 pts

Test continues on the other side of the sheet.

MSC 201 **25376**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

7. A rectangular density covers 0 to 4 and is zero elsewhere. What height makes total area 1?	A. 0.5. B. 0.25. C. 4. D. 1. 10 pts
8. Individual masses have SD 20 g; means of 25 independent masses have SD 4 g. Which spread applies to one package?	A. 4 g. B. 20 g. C. 0.8 g. D. 16 g. 10 pts
9. A file is 60 KB; its group has mean 50 KB and SD 5 KB. How far above the mean is it?	A. 2 standard deviations. B. 12 standard deviations. C. 0.5 standard deviations. D. 10 standard deviations. 10 pts
10. 40 of 100 sampled jobs succeed. What is the sample success proportion?	A. 0.60. B. 0.40. C. 40. D. 2.5. 10 pts

25376

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5. A.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

B.

10 pts

9. A.

10 pts

10.

B.

10 pts

MSC 201 **95118**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. 400 people are sampled without replacement from 20,000. Does this meet the at-most-10% sample-fraction guideline?	A. Yes; any random sample meets it. B. No; n must be below 30. C. Yes; the fraction is 2%. D. No; the fraction is 20%. 10 pts
2. Categories in a bar graph are reordered alphabetically. What changes?	A. The number of observations in each category. B. Every relative frequency. C. The most frequent category. D. The display order, but not category frequencies. 10 pts
3. A dataset's standard deviation is zero. What must be true?	A. All observations are zero. B. All observations are equal. C. The mean is zero. D. There is one high outlier. 10 pts
4. A box has 8 red and 2 blue tokens; 5 are drawn without replacement. What is the minimum red count?	A. 5. B. 3. C. 0. D. 2. 10 pts
5. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. They must have equal tails. B. The model with the larger SD. C. The model with the smaller SD. D. The one with more recorded observations regardless of SD. 10 pts
6. A sum distribution has mean 100 and SD 4. What total is 2 SD above its mean?	A. 102. B. 200. C. 108. D. 104. 10 pts
7. The numbers of pages in six jobs are 1, 2, 2, 4, 4, 5. Which statement about the mode is correct?	A. Only 4 is a mode. B. The mode is 3, halfway between 2 and 4. C. Both 2 and 4 are modes. D. Only 2 is a mode. 10 pts

Test continues on the other side of the sheet.

MSC 201 **95118**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A variable has mean 20 and SD 4. What z-score corresponds to 12?	A. 2. B. -8. C. 3. D. -2. 10 pts
9. The model changes from U(0,10) to U(0,20). What happens to P(X<5)?	A. It halves. B. It stays unchanged. C. It doubles. D. It becomes 1. 10 pts
10. A fair coin gives 8 heads in 10 tosses. What follows?	A. The coin's future head probability is now 0.8. B. Fairness requires exactly 5 heads in every ten tosses. C. The next two tosses must be tails. D. Such a short-run imbalance can occur with a fair coin. 10 pts

95118

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

C.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

D.

10 pts

9. A.

10 pts

10.

D.

10 pts

MSC 201 **81769**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A systematic sample starts at ID 3 and takes every tenth ID. Which IDs appear first?	A. 3, 6, 9. B. 3, 13, 23. C. 10, 20, 30. D. 3, 10, 20. 10 pts
2. A supplied probability table lists P(X=0),...,P(X=8). To find P(X<=2), which entries are needed?	A. The entries for 0, 1, and 2. B. The expected value. C. Only the entry for 2. D. The entries for 3 through 8. 10 pts
3. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A proportion interval regardless of variable type. B. A known-SD z interval treating s as known population SD. C. A Poisson count probability. D. A one-sample t interval. 10 pts
4. A supplied normal table gives P(Z<1.2)=0.8849. What is P(Z>1.2)?	A. 0.8849. B. 0.44245. C. 0.1151. D. 1.8849. 10 pts
5. Two servers share one power supply. Why might independent failure events be a poor model?	A. Different server names prove independence. B. Failure probabilities must sum to 1. C. Shared equipment guarantees no failures. D. A power failure can affect both servers together. 10 pts
6. Someone claims mean > median is a universal proof that any dataset is right-skewed. What is the best response?	A. It proves every observation exceeds the median. B. It guarantees exactly one mode. C. It proves the dataset is normal. D. It is a useful common pattern, not a universal characterization. 10 pts
7. Why report a confidence interval instead of only a sample mean?	A. It guarantees the true mean is the midpoint. B. It lists every individual observation. C. It removes sampling bias. D. It expresses uncertainty in estimating the population mean. 10 pts

Test continues on the other side of the sheet.

MSC 201 **81769**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

8. Which sequence gives the first success on trial 4?	A. Success, failure, failure, failure. B. Failure, failure, failure, success. C. Failure, success, failure, success. D. Four successes. 10 pts
9. Independent sample size increases. What happens to spread in original units?	A. Both rise. B. Both fall. C. SD of sums rises; SE of averages falls. D. Both stay unchanged. 10 pts
10. An exponential arrival rate doubles. What happens to mean waiting time?	A. It halves. B. It stays unchanged. C. It doubles. D. It becomes zero. 10 pts

81769

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

D.

10 pts

6.

D.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

C.

10 pts

10. A.

10 pts

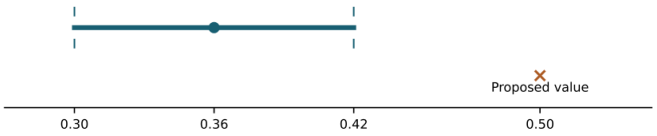
MSC 201 **15101**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. An exact continuous duration has P(X=4)=0. Can a timer report 4 seconds?	A. Only if the density equals zero everywhere. B. No; 4 seconds can never be recorded. C. Yes; rounding collects a positive-width interval of exact values. D. Only if the model becomes discrete before rounding. 10 pts
2. A population proportion interval is 0.30 to 0.42. How should a proposed proportion 0.50 be described? 	A. Compatible because 0.50 is a valid probability. B. Proved impossible under every future sample. C. Outside the interval and incompatible with this interval estimate. D. The required sample estimate. 10 pts
3. A stemplot includes an empty stem 2 between stems 1 and 3. What does that show?	A. The data were lost. B. All values are exactly 20. C. Zero is the only value. D. There are no values in the twenties. 10 pts
4. An independent fair coin has just produced five tails. What is the chance of heads next?	A. Greater than 1/2 because heads is due. B. Less than 1/2 because tails is favored. C. 1. D. 1/2. 10 pts
5. A game has negative expected net gain. Can a player still win money on one play?	A. No; every outcome must lose money. B. No; the largest prize must be negative. C. Yes, if positive-gain outcomes have positive probability. D. Only after the expected value changes. 10 pts
6. Only 5% exceed cutoff k. Which percentile is k?	A. 5th. B. 100th. C. 50th. D. 95th. 10 pts

Test continues on the other side of the sheet.

MSC 201 **15101**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

7. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The observations must include negative waits. B. It is approximation error outside the true support. C. The true sample mean can actually be below every observation. D. The normal model is exact at every tail. 10 pts
8. Q1 is 8 and Q3 is 14. What does the interval 8 to 14 describe approximately?	A. The lowest quarter only. B. The mean's confidence interval. C. The middle half of observations. D. Every observation. 10 pts
9. A model describes the mass of one package. Is its SD automatically appropriate for the mean mass of 100 independent packages?	A. No; the average has less spread. B. Yes; both are measured in grams. C. No; averages have no units. D. Yes; averages always vary more. 10 pts
10. A dataset has mean 8. One observation below 8 is removed. What happens to the mean of the remaining values?	A. It becomes the removed value. B. It increases. C. It always stays 8. D. It decreases. 10 pts

15101

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

C.

10 pts

9. A.

10 pts

10.

B.

10 pts

MSC 201 **20927**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A machine's success chance steadily falls as it warms. Which binomial requirement is questionable?	A. A fixed number of trials. B. A numerical success count. C. Constant success probability across trials. D. There being two possible outcomes. 10 pts
2. Population SD doubles while sample size stays fixed. What happens to SE?	A. It stays unchanged. B. It quadruples. C. It halves. D. It doubles. 10 pts
3. A is 6 above its mean with SD 3; B is 8 above its mean with SD 8. Which is more extreme above its group center?	A. They are equally extreme. B. The larger raw difference always determines extremeness. C. B. D. A. 10 pts
4. A population contains 4 observations with a sum of squared deviations from its mean of 36. What is the population variance?	A. 9. B. 12. C. 3. D. 36. 10 pts
5. A box has 3 red and 2 blue tokens. After a red token is drawn and not replaced, what happens to the red chance on the next draw?	A. It rises from 3/5 to 3/4. B. It stays 3/5. C. It becomes zero. D. It falls from 3/5 to 2/4. 10 pts
6. A stemplot key says stem 2, leaf 5 means 25 KB. Stem 3 has leaves 0, 2, 2. Which values does it show?	A. 30, 32, 32 KB. B. 3, 0, 2, 2 KB. C. 30, 32 KB only. D. 300, 320, 320 KB. 10 pts
7. A symmetric proportion interval is 0.20 to 0.30. What is the estimate and margin?	A. Estimate 0.25; margin 0.05. B. Estimate 0.20; margin 0.10. C. Estimate 0.30; margin 0.25. D. Estimate 0.10; margin 0.05. 10 pts

Test continues on the other side of the sheet.

MSC 201 **20927**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A sample contains 100 observations, but population SD is unknown. Does large n make that SD known?	A. No; it is still estimated from data. B. No; the sample mean must also be unknown. C. Yes; the sample SD becomes the population parameter by definition. D. Yes; n>=30 makes population SD exact. 10 pts
9. Under a constant errors-per-page Poisson model, document length doubles. What happens to the mean error count?	A. It halves. B. It doubles. C. It becomes the square root of the original mean. D. It stays unchanged. 10 pts
10. A U(0,10) sample has mean 5.4. Must the model be wrong?	A. Yes; all uniform values must equal 5. B. No; any mean proves the model. C. Yes; every sample must average exactly 5. D. No; a sample mean can differ from the model mean. 10 pts

20927

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4. A.

10 pts

5.

D.

10 pts

6. A.

10 pts

7. A.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **19351**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. An open voluntary poll has 50,000 replies and a very small calculated margin. What remains uncertain?	A. Whether 50,000 exceeds 30. B. Whether a proportion is numerical. C. Whether its respondents represent the target population. D. Whether the calculated margin must be negative. 10 pts
2. A fitted normal model predicts 2% above a cutoff, while a large observed dataset has 18% above it. What should be investigated?	A. Whether probabilities above 2% are impossible. B. Only the number of decimal places. C. Whether the model fits the observed tail. D. Whether observations should be changed to fit the model. 10 pts
3. A Poisson count has mean 9. What is its standard deviation?	A. 81. B. 3. C. 9. D. 0. 10 pts
4. Two samples have the same observed mean above a claimed population mean and the same population SD, but sizes 25 and 100. Which mean is farther above in SE units?	A. The size-100 mean. B. The size-25 mean. C. The larger sample changes the claimed mean. D. They are equally far. 10 pts
5. Tokens are replaced and mixed after each draw. Compared with leaving them out, what does replacement preserve?	A. The original color probabilities on each independent draw. B. A guarantee that colors alternate. C. The same token being selected every time. D. A decreasing total number of tokens. 10 pts
6. A rate is stated per hour but a cutoff is inserted as a number of minutes. What must be done before using the tail formula?	A. Round the probability to an integer. B. Express rate and cutoff in compatible time units. C. Replace the cutoff with the mean. D. Multiply both numbers by the sample size. 10 pts
7. Exact wait times are replaced by short, medium, and long categories. What information is lost?	A. The ordering of the three categories. B. The exact differences between wait times. C. Whether any observations exist. D. The number of observations. 10 pts

Test continues on the other side of the sheet.

MSC 201 **19351**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Two services have mean wait 10 minutes, but SDs 1 and 6 minutes. What does the larger SD suggest?	A. Waits vary more around the same mean. B. The mean must actually be 16 minutes. C. The median must be 6 minutes. D. Every wait is six minutes longer. 10 pts
9. Most files are small, with a few extremely large files. Which better describes a typical file's size?	A. The maximum. B. The median. C. The mean student ID. D. The total size alone. 10 pts
10. A student treats z=2 as a probability of 2. What is wrong?	A. A z-score can never exceed 1. B. A normal model cannot have positive z-scores. C. A z-score must be a percentage. D. A standardized distance is not an area probability. 10 pts

19351

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

B.

10 pts

4. A.

10 pts

5. A.

10 pts

6.

B.

10 pts

7.

B.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **59814**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. The slides recommend plus-four when confidence is at least 90% and n>=10. Does n=24 at 85% confidence meet that recommendation?	A. Yes; plus-four has no stated conditions. B. No; the confidence level is below the stated threshold. C. Yes; n alone is sufficient. D. No; n must exceed 100. 10 pts
2. A model allows counts 0,1,2,... without a finite upper limit. Can it still be discrete?	A. Yes; countably many separated values can be discrete. B. No; all infinite possibilities are continuous. C. No; a count must stop at 10. D. Only if every count has equal probability. 10 pts
3. Which design permits an exact binomial count model with independent randomized draws?	A. Drawing most of a small population without replacement. B. Replacing and remixing each drawn token. C. Drawing without replacement until the box is empty. D. Always stopping at first red. 10 pts
4. Only successful downloads are sampled to estimate duration for all attempts. Does a small SE remove that selection problem?	A. No; precision does not correct selection bias. B. No; SE must equal the mean. C. Yes; enough successful downloads represent failures automatically. D. Yes; a normal curve proves unbiased selection. 10 pts
5. A strongly skewed variable is converted to z-scores. Does it become normal?	A. No; z-scores cannot be negative. B. No; standardization shifts and rescales without removing skew. C. Yes; subtracting the mean removes the tail. D. Yes; every z-score distribution is standard normal. 10 pts
6. A grader does not know which practice format each paper's author used. What problem does this reduce?	A. All variation in student ability. B. Expectations influencing grading. C. The possibility of random assignment. D. Students being absent from the population. 10 pts

Test continues on the other side of the sheet.

MSC 201 **59814**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Which question targets a population proportion?	A. What is the median number of pages? B. What is the mean duration of jobs? C. What is the SD of file sizes? D. What fraction of jobs finish on time? 10 pts
8. A fixed fee increases by 2 dollars. What happens to net-gain mean and SD?	A. Both fall by 2. B. Mean falls by 2; SD is unchanged. C. Mean is unchanged; SD rises. D. Both double. 10 pts
9. A constant-rate process averages 0.5 requests per minute. What is the mean request count over 10 minutes?	A. 5. B. 20. C. 0.5. D. 10. 10 pts
10. A modified box plot marks 30 as a separate high outlier. Its upper whisker ends at 18. What does 18 represent?	A. The maximum of all observations. B. The largest observed value not flagged as an outlier. C. The mean. D. The upper fence whether observed or not. 10 pts

59814

TEST PAGE 1

1. B.

10 pts

2. A.

10 pts

3. B.

10 pts

4. A.

10 pts

5. B.

10 pts

6. B.

10 pts

TEST PAGE 2

7. D.

10 pts

8. B.

10 pts

9. A.

10 pts

10. B.

10 pts

MSC 201 **48376**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Compare 4, 5, 6 with 1, 5, 9. Which statement is correct?	A. The first has more spread. B. Equal means imply equal spread. C. The second has a larger mean. D. They have equal means, but the second has more spread. 10 pts
2. Someone claims mean > median is a universal proof that any dataset is right-skewed. What is the best response?	A. It guarantees exactly one mode. B. It proves the dataset is normal. C. It is a useful common pattern, not a universal characterization. D. It proves every observation exceeds the median. 10 pts
3. Which quantity describes how sample means vary from sample to sample?	A. Standard error. B. The sample size. C. The population mean. D. The range of all individuals alone. 10 pts
4. Independent trials have p=0.25. What is the expected number of trials through first success?	A. 0.25. B. 4. C. 3. D. 25. 10 pts
5. Which change best addresses a poll that includes only frequent library users?	A. Broaden selection to include nonusers. B. Collect ten times as many frequent-user responses. C. Shuffle the existing responses. D. Report more decimal places. 10 pts
6. A wait is uniform from 0 to 12 minutes. What is P(3<X<6)?	A. 3/12. B. 3/6. C. 9/12. D. 6/12. 10 pts
7. One model is symmetric and bell-shaped; another has a long right tail and decreases from zero onward. Which pairing fits those shapes?	A. Exponential first; normal second. B. Normal first; exponential second. C. Both must be uniform. D. Uniform first; normal second. 10 pts

Test continues on the other side of the sheet.

MSC 201 **48376**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Two mutually exclusive events each have positive probability. Can they be independent?	A. Only if their probabilities add to 1. B. Yes; disjoint always means independent. C. Yes; both have positive probabilities. D. No; one occurring makes the other impossible. 10 pts
9. For one normal model, P(X<40)=0.20 and P(X<60)=0.85. What is P(40<X<60)?	A. 1.05. B. 0.65. C. 0.85. D. 0.20. 10 pts
10. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 16. B. 15. C. 256. D. 17. 10 pts

48376

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3. A.

10 pts

4.

B.

10 pts

5. A.

10 pts

6. A.

10 pts

7.

B.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

B.

10 pts

10.

B.

10 pts

MSC 201 **82247**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A college wants each academic program represented in a survey. Which plan best achieves that directly?	A. Select the first students entering one room. B. Randomly sample students within every program. C. Survey only the largest program. D. Invite only students who previously complained. 10 pts
2. Compare 4, 5, 6 with 1, 5, 9. Which statement is correct?	A. Equal means imply equal spread. B. The first has more spread. C. They have equal means, but the second has more spread. D. The second has a larger mean. 10 pts
3. The success chance improves after each failed attempt because the process learns. Why is a geometric model questionable?	A. Failures are counted. B. The first success occurs eventually. C. There is a stopping rule. D. The success probability is not constant. 10 pts
4. For a standard normal variable, compare P(Z<-1.4) and P(Z>1.4).	A. The left tail is larger. B. They are equal. C. The right tail is larger. D. Both equal 0.5. 10 pts
5. Waits are nonnegative. Is that sufficient to establish an exponential model?	A. No; the arrival process and distributional assumptions still need support. B. Yes; every nonnegative variable is exponential. C. No; exponential values must be negative. D. Yes; nonnegative variables have equal mean and SD. 10 pts
6. A frequency table lists 1 page for 8 jobs and 5 pages for 2 jobs. Why is (1+5)/2 inappropriate for mean pages per job?	A. It counts too many individual jobs. B. It gives extra weight to 1-page jobs. C. It is always the median. D. It ignores how often each value occurs. 10 pts
7. What equation describes a continuous model's 90th percentile k?	A. P(X=k)=0.90. B. The mean equals 0.90k. C. P(X>k)=0.90. D. P(X<=k)=0.90. 10 pts

Test continues on the other side of the sheet.

MSC 201 **82247**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It uses the sample size. B. It assumes independent variation. C. It retains mass units. D. It predicts a nonnegative SD. 10 pts
9. Which expression computes P(X>0) for a nonnegative count X?	A. The mean of X. B. P(X=0). C. 1-P(X=1). D. 1-P(X=0). 10 pts
10. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It increases. B. It stays unchanged. C. It becomes zero. D. It decreases. 10 pts

82247

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5. A.

10 pts

6.

D.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

D.

10 pts

10. A.

10 pts

MSC 201 **53857**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A Venn summary has 30 in A, including 8 in both A and B. How many are in A only?	A. 38. B. 8. C. 22. D. 30. 10 pts
2. Which change best addresses a poll that includes only frequent library users?	A. Broaden selection to include nonusers. B. Report more decimal places. C. Shuffle the existing responses. D. Collect ten times as many frequent-user responses. 10 pts
3. Individual mean mass is 3 g. What is the expected total for 10 items?	A. 3 g. B. 30 g. C. 10 g. D. Exactly 30 g in every shipment. 10 pts
4. Sample SD is 6 minutes and n=9; sqrt(9)=3. What estimated SE is used?	A. 18 minutes. B. 2 minutes. C. 6 minutes. D. 3 minutes squared. 10 pts
5. A fitted normal model predicts 2% above a cutoff, while a large observed dataset has 18% above it. What should be investigated?	A. Whether probabilities above 2% are impossible. B. Whether the model fits the observed tail. C. Only the number of decimal places. D. Whether observations should be changed to fit the model. 10 pts
6. X counts failures in exactly 20 independent trials; Y counts arrivals during an hour with no fixed trial total. Which distinction matters?	A. X must be continuous. B. X fits a binomial setting; Y may fit a Poisson setting if its assumptions hold. C. Y must be geometric. D. Both must have the same maximum. 10 pts

Test continues on the other side of the sheet.

MSC 201 **53857**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A box plot flags an unusually large print job. Which conclusion is warranted?	A. The job must be fraudulent. B. The value must be deleted before any summary. C. The rest of the data must be normal. D. The job deserves investigation; the plot alone does not prove error. 10 pts
8. Compare 4, 5, 6 with 1, 5, 9. Which statement is correct?	A. The second has a larger mean. B. The first has more spread. C. Equal means imply equal spread. D. They have equal means, but the second has more spread. 10 pts
9. A model predicts 5% of items fall outside specifications. Among 200 items, how many are expected outside?	A. 5. B. Exactly 10 must occur. C. 40. D. 10. 10 pts
10. X is uniform from 2 to 10. What can be said about P(X>12)?	A. It is the density height at 12. B. It is zero. C. It is larger than P(X>8). D. It is 1/2. 10 pts

53857

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

D.

10 pts

9.

D.

10 pts

10.

B.

10 pts

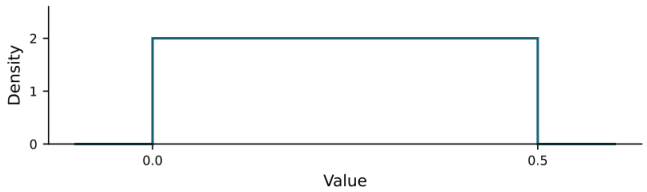
MSC 201 **50404**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A table contains 5 students using both devices. What does this show about mutual exclusivity?	A. They are mutually exclusive. B. The two use events are not mutually exclusive. C. Neither event has positive probability. D. They must be independent. 10 pts
2. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. Yes; its probability is 2 everywhere. B. Yes; every density height must be at most 1. C. No; its total area is 1, and height is not probability. D. No; total area may be any positive number. 10 pts
3. Why use planning proportion 0.5 when no prior estimate is available?	A. It maximizes p(1-p), giving a conservative required sample size. B. It guarantees half the population has the trait. C. It makes the sample unbiased automatically. D. It minimizes all margins to zero. 10 pts
4. Data are 0, 8, 9, 9, 10. Which center is lower?	A. The median. B. The mean. C. They are equal. D. The maximum. 10 pts
5. Individual mean mass is 3 g. What is the expected total for 10 items?	A. 3 g. B. 10 g. C. Exactly 30 g in every shipment. D. 30 g. 10 pts
6. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 10%. B. 45%. C. 90%. D. 5%. 10 pts

Test continues on the other side of the sheet.

MSC 201 **50404**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A selected student's name is removed before the next random draw. What changes?	A. Every remaining student becomes impossible to select. B. That student cannot be selected again. C. Selection has become a census. D. The same name must be selected next. 10 pts
8. A supplied Poisson table gives P(X=0)=0.12. How is the probability of at least one found?	A. 0.12. B. 0.12 squared. C. 1-0.12. D. The mean count. 10 pts
9. Which question targets a population proportion?	A. What fraction of jobs finish on time? B. What is the SD of file sizes? C. What is the mean duration of jobs? D. What is the median number of pages? 10 pts
10. Probability is shifted from a low outcome to a higher outcome, with other probabilities unchanged. What happens to expected value?	A. It decreases. B. It cannot change. C. It must become the largest outcome. D. It increases. 10 pts

50404

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

C.

10 pts

9. A.

10 pts

10.

D.

10 pts

MSC 201 **13695**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A symmetric confidence interval is 8 to 12. What point estimate is at its center?	A. 8. B. 12. C. 4. D. 10. 10 pts
2. A database codes printer brands as 1, 2, and 3. Why would averaging these codes be misleading?	A. Every numerical entry measures an amount. B. The numbers are labels; recoding them could change the mean. C. The largest code is always the most common brand. D. Brand codes cannot be counted. 10 pts
3. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 10%. B. 45%. C. 90%. D. 5%. 10 pts
4. Which sequence gives the first success on trial 4?	A. Success, failure, failure, failure. B. Four successes. C. Failure, failure, failure, success. D. Failure, success, failure, success. 10 pts
5. An exponential waiting time has mean 5 minutes. What rate per minute matches it?	A. 5. B. 0.2. C. 0.5. D. 25. 10 pts
6. Which interval approximates the discrete event X=45?	A. 44.5<Y<45.5. B. Y=45 only. C. Y>45.5. D. Y<44.5. 10 pts
7. P(A)=0.4, P(B)=0.5, and P(A AND B)=0.2. Which check supports independence?	A. The intersection is positive. B. 0.5 minus 0.4 equals 0.1. C. 0.4 times 0.5 equals 0.2. D. 0.4 plus 0.5 equals 0.9. 10 pts

Test continues on the other side of the sheet.

MSC 201 **13695**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Most files are small, with a few extremely large files. Which better describes a typical file's size?	A. The mean student ID. B. The maximum. C. The total size alone. D. The median. 10 pts
9. A box plot flags an unusually large print job. Which conclusion is warranted?	A. The job must be fraudulent. B. The rest of the data must be normal. C. The job deserves investigation; the plot alone does not prove error. D. The value must be deleted before any summary. 10 pts
10. A normal model assigns 0.90 inside an accepted interval. What probability is outside it?	A. 0.10. B. 0.05 necessarily. C. 0.90. D. 1.90. 10 pts

13695

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6. A.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **57826**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A proportion interval is 20% to 30%. Which conclusion matches the target?	A. Exactly 25% of the sample must be successful. B. Every individual has a value between 20 and 30. C. The population percentage is estimated between 20% and 30%. D. The mean duration is between 20 and 30 minutes. 10 pts
2. Two datasets have identical five-number summaries. What follows?	A. They must contain identical observations. B. Their sample sizes must be identical. C. Their means must be identical. D. They can still have different internal distributions. 10 pts
3. Histogram bins have unequal widths but equal counts. Why can equal raw-count bar heights mislead about concentration?	A. Wider bins always contain smaller values. B. Every histogram must be a time series. C. The wider bin spreads its count over more numerical space. D. Equal counts require equal widths. 10 pts
4. A wait is uniform from 0 to 12 minutes. What is $P(3<X<6)$?	A. 3/6. B. 9/12. C. 6/12. D. 3/12. 10 pts
5. Two successes in four trials can occur in several orders. Why must all those orders be considered?	A. Only the first successful order can occur. B. Different orders are the same elementary outcome. C. The count event includes every order with two successes. D. The expected count is an order. 10 pts
6. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 99.7%. B. 3%. C. 0.15%. D. 0.3%. 10 pts

Test continues on the other side of the sheet.

MSC 201 **57826**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. SE is 4 at n=25. What is SE at n=100, with the same population SD and independence?	A. 2. B. 1. C. 4. D. 8. 10 pts
8. A bell-shaped distribution of scores has mean 50 and standard deviation 5. Approximately what percentage of scores lies between 45 and 55?	A. 68%. B. 95%. C. 99.7%. D. 50%. 10 pts
9. A normal model's SD doubles with mean fixed. What happens to its model IQR?	A. It stays unchanged. B. It becomes zero. C. It quadruples. D. It doubles. 10 pts
10. Among 200 recorded jobs, 30 were late. What does 30/200 describe?	A. A guarantee that the next job is late. B. The number of late jobs that must occur tomorrow. C. The exact long-run probability under every possible condition. D. The observed late-job relative frequency. 10 pts

57826

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6.

D.

10 pts

TEST PAGE 2

7. A.

10 pts

8. A.

10 pts

9.

D.

10 pts

10.

D.

10 pts

MSC 201 **21950**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A normal model has mean 50 and SD 2. Which interval contains about 99.7%?	A. 44 to 56. B. 46 to 54. C. 50 to 56. D. 48 to 52. 10 pts
2. X is uniform from 0 to 12. Given X>4, what is P(X>8)?	A. 8/12. B. 8/8. C. 4/8. D. 4/12. 10 pts
3. Categories in a bar graph are reordered alphabetically. What changes?	A. Every relative frequency. B. The number of observations in each category. C. The display order, but not category frequencies. D. The most frequent category. 10 pts
4. What does a 95% confidence procedure mean in repeated appropriate sampling?	A. The sample mean equals the population mean 95% of the time. B. 95% of observations lie in every interval. C. About 95% of intervals constructed this way contain the fixed population mean in the long run. D. Every interval has 95% of its length below the mean. 10 pts
5. A box has 8 red and 2 blue tokens; 5 are drawn without replacement. What is the minimum red count?	A. 5. B. 0. C. 2. D. 3. 10 pts
6. A file is 60 KB; its group has mean 50 KB and SD 5 KB. How far above the mean is it?	A. 12 standard deviations. B. 0.5 standard deviations. C. 10 standard deviations. D. 2 standard deviations. 10 pts
7. For a normal distribution, how do Q1 and Q3 lie relative to the mean?	A. Both lie above it. B. Both equal it. C. Q3 must be twice Q1. D. They are equally distant on opposite sides. 10 pts

Test continues on the other side of the sheet.

MSC 201 **21950**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

8. Most files are small, with a few extremely large files. Which better describes a typical file's size?	A. The mean student ID. B. The maximum. C. The median. D. The total size alone. 10 pts
9. Which event is the complement of at least one failure in three trials?	A. At least two failures. B. No failures. C. Exactly one failure. D. All three failures. 10 pts
10. A population has mean 50. Where are unbiased random-sample means centered?	A. At the population SD. B. At 50. C. At zero. D. At 50 divided by sample size. 10 pts

21950

TEST PAGE 1

1. A.

10 pts

2. C.

10 pts

3. C.

10 pts

4. C.

10 pts

5. D.

10 pts

6. D.

10 pts

7. D.

10 pts

TEST PAGE 2

8. C.

10 pts

9. B.

10 pts

10. B.

10 pts

MSC 201 **37932**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A mean interval is 10 to 14 minutes. Which report matches its target?	A. Exactly 95 of the next 100 durations will lie there. B. The procedure estimates the population mean duration between 10 and 14 minutes. C. The median is guaranteed to equal 12. D. Every future duration will be between 10 and 14. 10 pts
2. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 0.3%. B. 99.7%. C. 0.15%. D. 3%. 10 pts
3. A uniform interval centered at 10 doubles its width while keeping center 10. What happens?	A. Mean doubles; SD stays unchanged. B. Both stay unchanged. C. Mean stays 10; SD doubles. D. Both double. 10 pts
4. A systematic sample starts at ID 3 and takes every tenth ID. Which IDs appear first?	A. 3, 10, 20. B. 10, 20, 30. C. 3, 6, 9. D. 3, 13, 23. 10 pts
5. A Poisson count has mean 9. What is its standard deviation?	A. 0. B. 9. C. 81. D. 3. 10 pts
6. A population has 2 faulty cables among 10; 8 are sampled without replacement. What sample information constrains the count?	A. Only the population mean. B. Only whether the count is odd. C. Only the first cable's color. D. Both available faulty cables and available nonfaulty cables. 10 pts
7. From the same sample, a 90% and 99% mean interval are built with the same method. Which statement fits?	A. Both widths must be identical. B. The 99% interval is wider but has the same center. C. The 99% interval has a different sample mean. D. The 90% interval is wider. 10 pts

Test continues on the other side of the sheet.

MSC 201 **37932**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

8. A service meets its deadline with probability 0.80. What is the probability of missing it?	A. 1.80. B. 0.20. C. 0.40. D. 0.80. 10 pts
9. Sample size rises from 25 to 100 under unbiased independent sampling. What happens to the center of sample means?	A. It stays at the population mean. B. It quadruples. C. It halves. D. It becomes zero. 10 pts
10. Most files are small, with a few extremely large files. Which better describes a typical file's size?	A. The mean student ID. B. The total size alone. C. The maximum. D. The median. 10 pts

37932

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

D.

10 pts

7.

B.

10 pts

TEST PAGE 2

8.

B.

10 pts

9. A.

10 pts

10.

D.

10 pts